BIOINFO
GIW ISCB-Asia
Seoul · Nov 17–20, 2026

Predictability Is Not Substitutability

A cost-of-substitution framework for H&E-based molecular prediction across 5 cancers

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P082
Long abstract
doi.org/10.5281/zenodo.24068874

Summary. Discrimination (AUROC) shows that a molecular marker is *predictable* from H&E; it does not show that the prediction can *substitute* for the molecular test. Under one protocol, 10 sealed-forward endpoints in lung, gastric and head-and-neck cancer resolved as **7 candidates (1 confirmed, 6 undecided)** and **3 positive controls**. A near-ceiling AUROC (0.939, lung histology) was excluded by a site audit; the six unconfirmed candidates were undecided rather than negative.

1. Background

H&E models predict MSI, driver mutations and subtypes at high reported AUROC [1–3], but AUROC does not say which treatment a patient receives when the prediction is wrong [4].

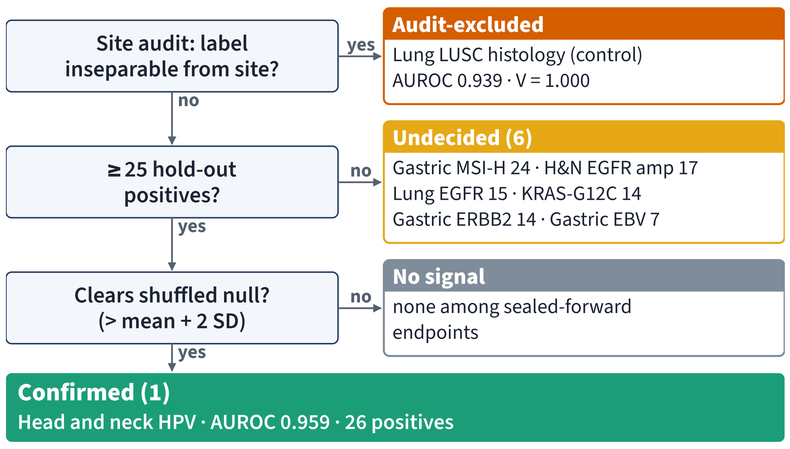
In **our prior work**, an H&E recurrence model reached sensitivity 0.857 for low-risk but 0.529 for high-risk patients [5]. Such predictors are often confounded [6]; in a 33-task benchmark, mean AUROC fell from 0.689 to 0.630 externally [7].

2. Cost of substitution

Under a routing fixed in advance, the cost for reference axis a is the mean treatment distance d (0–1) between reference and routed axis over its n_a slides:

$$cost_a = (1/n_a)\ \Sigma_{i:\ a_i=a}\ d(a,\ \hat{a}_i)$$

The **primary measure** is the distance-free **misassignment rate**. It prices an error at the moment of assignment, not drug response.



Controls: H&N grade passed; gastric Lauren failed (95% CI includes 0.5); lung histology above.

Verdict rules. Counts are the 10 sealed-forward endpoints of Figure 2. Gastric ERBB2 is classified Undecided.

3. Methods

- **Cohorts:** breast (TCGA-BRCA anchor, ~1,010 slides), lung, colorectal, gastric, head and neck.
- **Pre-processing:** 256 × 256 tiles at 20×, Otsu segmentation, ≤ 5,000 tiles/slide.
- **Model:** UNI v1 embeddings (1,024-d) [10] + CLAM-SB attention MIL [11].
- **Evaluation:** site-disjoint hold-out only.
- **Controls:** 5-seed label-shuffled null (AUROC > null mean + 2 SD); prevalence baseline; histology-only / pixel-mean baseline (reported, not a verdict criterion); 1,000× bootstrap CI.
- Predictions **sealed by commit** before hold-out results were produced (lung, gastric, head and neck); the V = 1 audit-exclusion rule was formalized after the site audit.
- **Scope:** 10 sealed-forward endpoints (incl. 3 positive controls); breast/colorectal (4) and endometrial (1) reported but not counted.

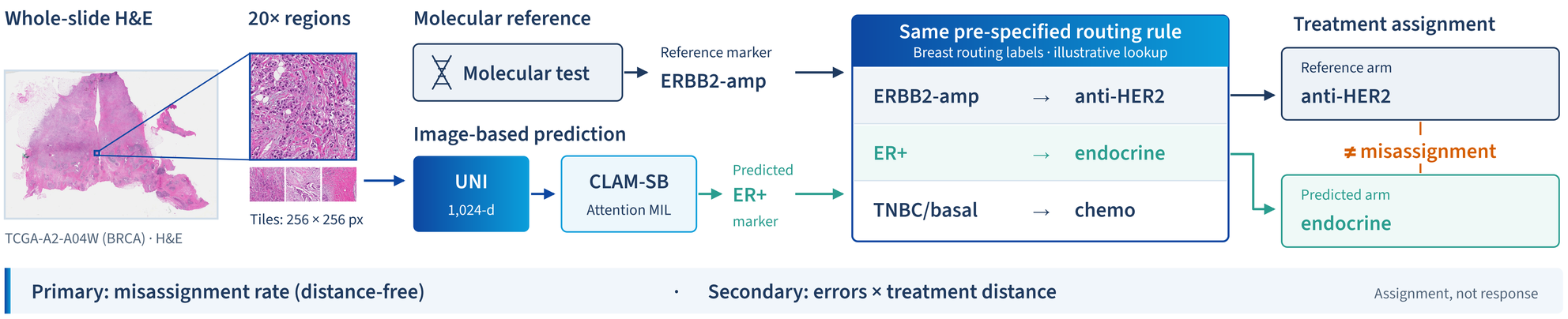


Figure 1. Framework. The molecular result and the H&E prediction pass through the same pre-specified treatment routing; a disagreement in the assigned arm is a misassignment. Breast-anchor misassignment: see Figure 5.

4. Results

Table 1. Selected sealed-forward hold-out endpoints: lung, gastric, head and neck.

Endpoint	AUROC	95% CI	Hold-out pos.	Verdict
Head and neck HPV	0.959	0.921–0.986	26	Confirmed‡
Lung LUSC histology (control)	0.939	0.905–0.967	153	Audit-excluded
Gastric MSI-H	0.860	0.759–0.941	24	Undecided
Lung EGFR activating	0.852	0.725–0.953	15	Undecided
Lung KRAS-G12C	0.681	0.577–0.783	14	Undecided
Gastric ERBB2 amplification	0.644	0.523–0.771	14	Undecided

Also undecided (Figure 2): H&N EGFR amp (17 pos.), gastric EBV (7). Not counted (anchor/retrospective): breast HER2 0.599; colorectal anti-EGFR 0.705 (84 pos.), MSI-high 0.918 (21), BRAF V600E 0.882 (15). ‡ Primary model (UNI). With 20 label shuffles HPV cleared the null under UNI, UNI2-h and Virchow2 [12] (5 shuffles: Virchow2 0.9199 vs null threshold 0.9234). HPV pixel-mean baseline AUROC 0.922 [0.854–0.975].

Confirmation. Head and neck HPV was the only non-control endpoint meeting the primary rule. It has known histological correlates (non-keratinising, basaloid morphology) [8] and HPV association has been predicted from H&E in two cohorts of 273 patients [8].

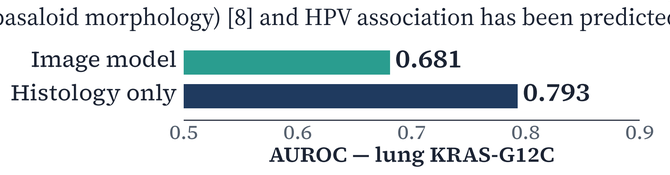


Figure 4. Baseline check: histological subtype alone, with no image, beats the image model for lung KRAS-G12C. Remains Undecided (14 positives): this baseline is a reported control, not a verdict rule, so the contrast is an observation only.

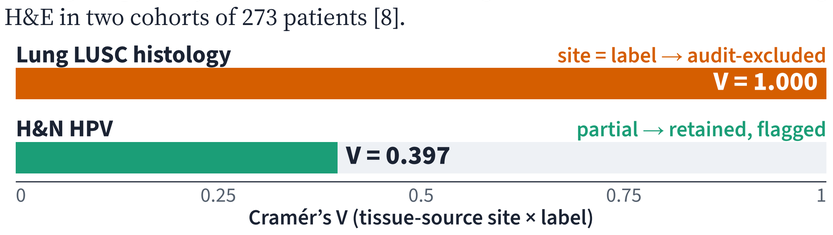


Figure 3. Site-confounding audit. TCGA institution codes coincide exactly with LUAD/LUSC status [9], so the 0.939 AUROC was audit-excluded; HPV shows a partial association.

4a. Positive-count ceiling

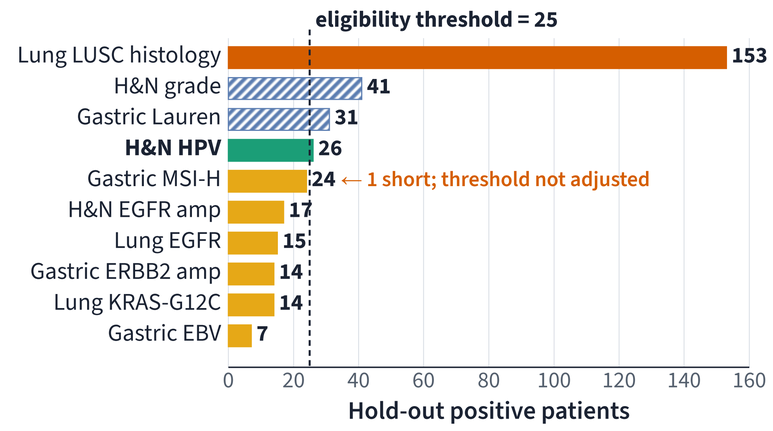


Figure 2. Hold-out positives vs the eligibility threshold of 25 (hatched: positive controls; lung histology, also a control, shown as audit-excluded). All six non-HPV candidate endpoints fall below it — undecided, not negative. Gastric MSI-H fell one short; the criterion was not adjusted.

4b. Breast anchor

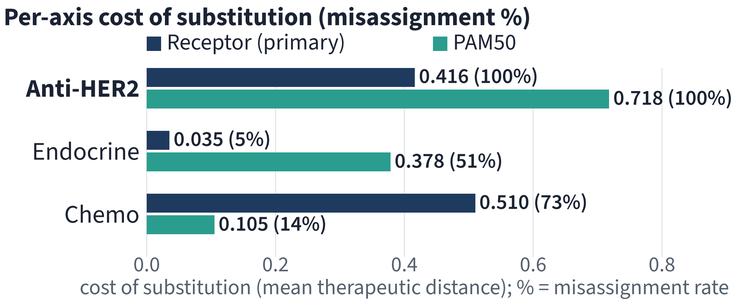


Figure 5. Per-axis substitution cost under two pre-specified routings. In external CPTAC-BRCA, every anti-HER2 slide is misassigned under both; endocrine/chemotherapy costs reverse between routings (no safety claim). The anti-HER2 − endocrine contrast excludes 0 under both (receptor 0.381 [0.331–0.427]; PAM50 0.340 [0.251–0.425]). In an external pCR cohort (exploratory), the anti-HER2 prediction did not stratify response (AUROC 0.533).

5. Discussion

Retrospective and hypothesis-generating. Site-disjoint splits block leakage, not confounding. Positive count alone does not explain the pattern (Figure 6).

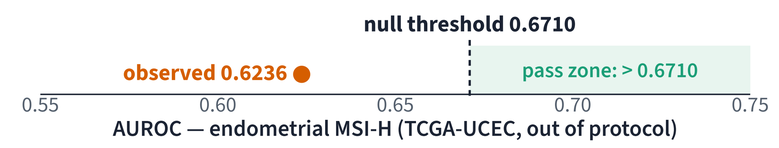


Figure 6. Eligibility alone is not enough. Out-of-protocol endometrial MSI-H met the 25-positive criterion yet failed null exclusion; excluded from the confirmation tally.

6. Conclusion

1. Explicit verdict rules separated **1 confirmed** and **6 undecided** candidates, and audit-excluded a near-ceiling control.
2. The framework rejected a near-ceiling AUROC and, under the primary model, retained one literature-supported endpoint.
3. Reporting the undecidable as **undecidable**, not negative, is the central point; the procedure transfers to any proposed H&E surrogate.

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Acknowledgement. This research was supported by the "Open Source AI·SW Developer and Community Support Program" funded by the Ministry of Science and ICT (MSIT), Republic of Korea. Conducted as an independent Pseudo Lab project; corporate affiliations are listed for identification only.

Code & results